

Lateral Canthotomy

Section 1: Case Summary

Scenario Title:	
Keywords:	Lateral Canthotomy
Brief Description of Case:	A 25 y/o M pitching in a Sunday baseball game is hit in the face by a line drive. He is brought to the ED by his friends complaining of decreased visual acuity to his right eye accompanied by significant right peri-orbital swelling. At triage his VA is OD 20/100 OS 20/25 but at the time of assessment VA OD is limited to detection of light and his pupil is fixed and dilated, extra-ocular movements are intact. The team should recognize the need for lateral canthotomy based on the history and physical exam findings (including IOP) and mobilize the appropriate resources for bed-side lateral canthotomy as well as the need for emergent ophthalmology consult.

Goals and Objectives	
Educational Goal:	To expose learners to the indications, contraindications, and technique of lateral canthotomy.
Objectives: (Medical and CRM)	Medical Objectives: 1. Describe the indications and contraindications for lateral canthotomy in the ED. 2. Demonstrate proper technique for the procedure including knowledge of anatomical landmarks, necessary equipment, local anesthetic application, and post-procedure care. CRM Objectives: 1. Communicate effectively with care team and specialist consultants.
EPAs Assessed:	Core EPA: Advanced procedures

Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☒ Senior Learners		☒ Staff
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional	
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1-2				
	Sim Actors: 1 Nurse.				
	Sim Techs: 1				

Scenario Development	
Date of Development:	25/2/21
Scenario Developer(s):	Drs. Stephen Watson, Jared Baylis, Lee Graham
Affiliations/Institutions(s):	The University of British Columbia
Contact E-mail:	sfwatson95@gmail.com
Last Revision Date:	
Revised By:	
Version Number:	



Lateral Canthotomy

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Randy Johnson		Age: 25		Gender: M	Weight: 95kg
Presenting complaint: Right orbital trauma					
Temp: 36.5	HR: 102	BP: 134/96	RR: 22	O ₂ Sat: 100%	FiO ₂ : RA
Cap glucose: 6.3 mmol/L			GCS: 15		
Triage note: Struck by baseball to right eye. No LOC, no N/V, A+O x 3.					
Allergies: Nil					
Past Medical History: Nil.			Current Medications: Nil.		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
No retrograde amnesia, not on any blood thinners, no c-spine concerns, no other injuries. Not intoxicated.	

B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal	Neuro: GCS 15, A+Ox3, moving x4 w/o any strength or sensory deficits. Ambulated into ED. Right pupil fixed and dilated at 7mm. Full EOM, some discomfort to right eye w/ EOM. VA OD= light detection. VA OS =20/25.
Resp: Normal	Head & Neck: Significant right orbital hematoma, proptosis of right eye w/ bloody chemosis, no appreciable step-offs or deformities to facial bones.
Abdo: Normal	MSK/skin: abrasions + hematoma to R orbit.
Other:	



Lateral Canthotomy

Section 3: Technical Requirements/Room Vision

A. Patient
☐ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☒ Task Trainer
☐ Hybrid
B. Special Equipment Required
Porcine model 2x hemostat Iris scissors 27G needle + 10mL syringe for local anesthetic administration Fenestrated drapes 22G + 10mL syringe for introduction of saline into the model orbit Tono-pen + covers for IOP measurement
C. Required Medications
Local anesthetic of choice.
D. Moulage
Induced chemosis w/ saline injection retro-orbitally.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i>

Lateral Canthotomy

Section 4: Actors and Standardized Patients

Actors and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Nurse	Can assist w/ securing supplies, drawing up local.

Lateral Canthotomy

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: NSR HR: 102 BP: 134/96 RR: 22 O ₂ SAT: 100% T: 36.5°C GCS: 15	<i>Patient alert, co-operative but complains of pain and loss of vision to right eye.</i>	<u>Expected Learner Actions</u> ☐ Recognize deteriorating visual acuity and potential for orbital compartment syndrome ☐ Call out for emergent ophthalmology consult ☐ Assemble required equipment for lateral canthotomy ☐ Consider CT head/facial bones post-procedure ☐ Update tetanus toxoid vaccination status ☐ Provide analgesia while assembling supplies		<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> -Deferral of lateral canthotomy → permanent vision loss <u>Triggers</u> <i>For progression to next state</i> Successful Lateral Canthotomy
2. Post-procedure		<u>Expected Learner Actions</u> ☐ Liaise with ophthalmology for follow up ☐ Update patient regarding plans ☐ Recheck intraocular pressure to ensure resolution of orbital compartment syndrome		<u>Modifiers</u> - - - <u>Triggers</u> - Action items complete → End case



Lateral Canthotomy

Appendix A: Laboratory Results

Not applicable



Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

Lateral Canthotomy

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

See associated write up with details of lateral canthotomy procedure

References

1. Gordon, S., & Macrae, H. (1950). Monocular Blindness As A Complication Of The Treatment Of A Malar Fracture. *Plastic and Reconstructive Surgery*, 6(3), 228-232. doi:10.1097/00006534-195009000-00005
2. Bailey, W. K., Kuo, P. C., & Evans, L. S. (1993). Diagnosis and treatment of retrobulbar hemorrhage. *Journal of Oral and Maxillofacial Surgery*, 51(7), 780-782. doi:10.1016/s0278-2391(10)80422-7
3. Popat, H., Doyle, P., & Davies, S. (2007). Blindness following retrobulbar haemorrhage—It can be



Lateral Canthotomy

prevented. *British Journal of Oral and Maxillofacial Surgery*, 45(2), 163-164.
doi:10.1016/j.bjoms.2005.06.028

4. Credland, T. (2015). 0173 The development of a model to simulate an intra ocular bleed and lateral canthotomy. *Technical Innovations/tech Room Topics*. doi:10.1136/bmjstel-2015-000075.15
5. Patel, S. R., Mishall, P., & Barmettler, A. (2019). A human cadaveric model for effective instruction of lateral canthotomy and cantholysis. *Orbit*, 39(2), 87-92. doi:10.1080/01676830.2019.1600151
6. Rowh, A. D., Ufberg, J. W., Chan, T. C., & Vilke, G. M. (2015). Lateral Canthotomy and Cantholysis: Emergency Management of Orbital Compartment Syndrome. *The Journal of Emergency Medicine*, 48(3), 325-330.
7. Mahon, B. M., & Desai, B. K. (2016). Lateral Canthotomy. *Atlas of Emergency Medicine Procedures*, 305-308. doi:10.1007/978-1-4939-2507-0_52

